Assignment-3: Machine Learning

MCA | Questions and Answers (1-10)

Question 1: Supervised Learning vs Unsupervised Learning (5 marks)

Question: *Define Supervised Learning. Explain the key difference between supervised learning and unsupervised learning with one example of each.*

Answer:

Supervised Learning is a machine learning approach in which a model is trained on a *labelled* dataset, i.e. every input x has a known output (label) y. The model learns a mapping f: X → Y so that it can predict the output for new, unseen inputs.

Basis	Supervised Learning	Unsupervised Learning
Data	Labelled (input + output)	Unlabelled (input only)
Goal	Predict a known target	Discover hidden structure/patterns
Feedback	Error against true labels guides learning	No explicit feedback
Tasks	Classification, Regression	Clustering, dimensionality reduction, association
Example	Spam detection: emails labelled spam / not spam are used to train a classifier	Customer segmentation: K-Means groups customers by purchase behaviour without labels

Question 2: Types of Supervised Learning Problems (5 marks)

Question: *What are the two main types of supervised learning problems? Clearly distinguish between them and give one real-world example for each type.*

Answer:

Basis	Classification	Regression
Output	Discrete class label / category	Continuous numerical value
Goal	Assign input to a class	Predict a quantity
Algorithms	Logistic Regression, Decision Tree, SVM, kNN	Linear Regression, Ridge/Lasso, Regression Trees
Metrics	Accuracy, Precision, Recall, F1	MSE, RMSE, MAE, R^2
Real-world example	Classifying a tumour as benign or malignant	Predicting house price from area, location and rooms

Question 3: Linear Regression (5 marks)

Question: *Explain the concept of Linear Regression. Write the equation of a simple linear regression model and define the terms (dependent variable, independent variable, intercept, and slope). What is the role of the residual/error term?*

Answer:

Linear Regression models the relationship between a dependent variable and one or more independent variables by fitting a straight line (or hyperplane) that best describes the data. The best-fit line is usually found by **Ordinary Least Squares (OLS)**, which minimises the sum of squared

errors.

Simple linear regression equation:

$$y = \beta_0 + \beta_1 x + \varepsilon$$

- **Dependent variable (y):** the target/response being predicted (e.g. salary).
- **Independent variable (x):** the input/predictor used for prediction (e.g. years of experience).
- **Intercept (β_0):** value of y when x = 0; where the line crosses the y-axis.
- **Slope (β_1):** change in y for a one-unit change in x. Estimated as $\beta_1 = \Sigma(x_i - x\blacksquare)(y_i - y\blacksquare) / \Sigma(x_i - x\blacksquare)^2$, and $\beta_0 = y\blacksquare - \beta_1 x\blacksquare$.

Role of the residual/error term (ε): The residual $e_i = y_i - \blacksquare_i$ is the difference between the actual and predicted value. It captures variation not explained by the model (noise, omitted variables, measurement error). OLS minimises SSE = Σe_i^2. Residuals are also used to check model assumptions (linearity, constant variance, normality, independence).

Question 4: Logistic Regression (5 marks)

Question: *How does Logistic Regression differ from Linear Regression? Write the logistic (sigmoid) function and explain why it is suitable for binary classification. What is the decision boundary in logistic regression?*

Answer:

Basis	Linear Regression	Logistic Regression
Problem type	Regression (continuous output)	Classification (probability of a class)
Output range	-∞ to +∞	0 to 1
Function	Straight line $y = \beta_0 + \beta_1 x$	Sigmoid applied to linear function
Loss	Mean squared error	Log loss (cross-entropy)
Estimation	Least squares	Maximum likelihood

Sigmoid (logistic) function:

$$\sigma(z) = 1 / (1 + e^{-z}), \quad \text{where } z = \beta_0 + \beta_1 x_1 + \ldots + \beta_n x_n$$

Why suitable for binary classification: the sigmoid squashes any real number into (0, 1), so the output can be interpreted as the probability P(y=1|x). It is smooth, differentiable, and gives an S-shaped curve with a clear transition between the classes. A threshold (usually 0.5) converts the probability to a class label.

Decision boundary: the surface where P(y=1|x) = 0.5, i.e. where z = 0. So the boundary is $\beta_0 + \beta_1 x_1 + \ldots + \beta_n x_n = 0$, which is linear (a line in 2-D, a hyperplane in higher dimensions). Points with z > 0 are predicted class 1, and z < 0 class 0.

Question 5: Decision Trees (5 marks)

Question: *Describe how a Decision Tree works for classification. Explain the concepts of Information Gain and Gini Impurity. Why can decision trees overfit the training data, and how can this problem be reduced?*

Answer:

Working: A decision tree repeatedly splits the dataset using feature tests. The *root node* holds all data; at each *internal node* the feature and threshold giving the best split (purest child nodes) is chosen; *leaf nodes* give the final class (majority class of samples reaching it). To classify a new sample, it is passed from root to a leaf by answering the tests along the path. Splitting stops when nodes are pure or a stopping rule (max depth, min samples) is met.

Entropy and Information Gain (used by ID3/C4.5):

$$Entropy(S) = -\Sigma\, p_i \log_2 p_i \qquad IG(S, A) = Entropy(S) - \Sigma\, (|S_v|/|S|)\, Entropy(S_v)$$

Information Gain is the reduction in entropy after splitting on attribute A; the attribute with the highest gain is chosen.

Gini Impurity (used by CART):

$$Gini(S) = 1 - \Sigma\, p_i^2$$

It is the probability of misclassifying a randomly chosen sample if labelled by the class distribution. It is 0 for a pure node (and 0.5 maximum for two balanced classes). The split with the lowest weighted Gini is chosen.

Why overfitting occurs: an unrestricted tree keeps splitting until each leaf is pure, so it memorises noise and outliers in the training data, giving low training error but poor test performance (high variance).

Reducing overfitting:

- Pre-pruning: limit max depth, minimum samples per split/leaf, or minimum gain.
- Post-pruning: grow the full tree, then remove branches that do not improve validation accuracy (e.g. cost-complexity pruning).
- Use ensembles such as Random Forest; use cross-validation and more training data.

Question 6: Ensemble Methods (5 marks)

Question: *What are Ensemble Methods? Differentiate between Bagging and Boosting with suitable examples of algorithms that use each technique. Why do ensemble methods generally perform better than a single model?*

Answer:

Ensemble methods combine the predictions of multiple base models (weak learners) to produce a single, stronger model.

Basis	Bagging (Bootstrap Aggregating)	Boosting
Training	Models trained **in parallel** on random bootstrap samples (with replacement)	Models trained **sequentially**; each corrects errors of the previous
Sample weighting	All samples equal	Misclassified samples get more weight/focus
Combining	Majority vote / average	Weighted vote / weighted sum
Main effect	Reduces **variance**	Reduces **bias** (and variance)
Risk	Less prone to overfitting	Can overfit on noisy data
Algorithms	Random Forest, Bagged Trees	AdaBoost, Gradient Boosting, XGBoost, LightGBM

Why ensembles perform better: individual models make different errors; when their errors are not perfectly correlated, averaging/voting cancels out many of them. This reduces variance (bagging) or bias (boosting), improves generalisation and robustness to noise, and gives higher accuracy than a single model.

Question 7: Support Vector Machines (5 marks)

Question: *Explain the working principle of Support Vector Machines (SVM) for binary classification. What are support vectors and the margin? Briefly describe the role of the kernel trick in SVM.*

Answer:

Working principle: SVM finds the **optimal separating hyperplane** $w \cdot x + b = 0$ that separates two classes with the **maximum margin**. Labels are $y \in \{-1, +1\}$ and the constraint is $y_i(w \cdot x_i + b) \geq 1$. The margin width is $2/||w||$, so maximising it is equivalent to:

$$\textit{minimise } (1/2)||w||^2 \quad \textit{subject to } y_i(w \cdot x_i + b) \geq 1$$

For non-separable data, a soft margin with slack variables and a penalty parameter C is used (large C = fewer misclassifications, small margin; small C = wider margin, more tolerance).

- **Support vectors:** the training points closest to the hyperplane, lying on the margin boundaries. They alone define the hyperplane; removing other points does not change it.
- **Margin:** the distance between the hyperplane and the nearest points of either class. A larger margin gives better generalisation.

Kernel trick: when data is not linearly separable, it is implicitly mapped to a higher-dimensional space where a linear separator exists. The kernel function $K(x_i, x_j)$ computes dot products in that space without explicitly performing the transformation, saving computation. Common kernels: Linear, Polynomial $(x \cdot y + c)^d$, RBF/Gaussian $\exp(-\gamma||x - y||^2)$, and Sigmoid.

Question 8: Classification Evaluation Metrics (5 marks)

Question: *List and briefly explain any four evaluation metrics used for classification problems (e.g., Accuracy, Precision, Recall, F1-Score, ROC-AUC). When is Accuracy not a good metric, and which metric would you prefer instead?*

Answer:

Based on the confusion matrix with TP, TN, FP, FN:

Metric	Formula	Meaning
Accuracy	(TP + TN) / (TP + TN + FP + FN)	Fraction of all predictions that are correct
Precision	TP / (TP + FP)	Of predicted positives, how many are truly positive (cost of false alarms)
Recall (Sensitivity)	TP / (TP + FN)	Of actual positives, how many were found (cost of missed cases)
F1-Score	2 × (Precision × Recall) / (Precision + Recall)	Harmonic mean of precision and recall; balances both
ROC-AUC	Area under TPR vs FPR curve	Ability to separate classes across all thresholds; 0.5 = random, 1 = perfect

When accuracy is not good: on **imbalanced datasets**. For example, if only 1% of transactions are fraud, a model that predicts 'not fraud' always achieves 99% accuracy but detects no fraud at all. In

such cases prefer **Recall** (when missing positives is costly, e.g. disease or fraud detection), **Precision** (when false alarms are costly, e.g. spam filtering), or the **F1-Score** / PR-AUC to balance both.

Question 9: Regression Evaluation Metrics (5 marks)

Question: *For regression problems, explain the following evaluation metrics: (a) Mean Squared Error (MSE) (b) Root Mean Squared Error (RMSE) (c) Mean Absolute Error (MAE) (d) R-squared (Coefficient of Determination). Which metric is more sensitive to outliers and why?*

Answer:

Let y_i be actual values, $\blacksquare_i$ predicted values, $y\blacksquare$ the mean of y, and n the number of samples.

- **(a) MSE** = $(1/n)\ \Sigma(y_i - \blacksquare_i)^2$. Average of squared errors; always non-negative, in squared units; penalises large errors heavily. Lower is better.
- **(b) RMSE** = $\sqrt{}$MSE. Square root of MSE, so it is in the same unit as the target and easier to interpret; still penalises large errors.
- **(c) MAE** = $(1/n)\ \Sigma|y_i - \blacksquare_i|$. Average absolute error; same unit as target; treats all errors linearly, so it is more robust.
- **(d) R-squared** = $1 - SS_{res}/SS_{tot} = 1 - \Sigma(y_i - \blacksquare_i)^2 / \Sigma(y_i - y\blacksquare)^2$. Proportion of variance in y explained by the model. Value 1 is perfect, 0 means no better than predicting the mean, and it can be negative for very poor models. (Adjusted R^2 accounts for the number of predictors.)

Most sensitive to outliers: MSE (and hence RMSE), because errors are **squared**. An error of 10 contributes 100, while an error of 1 contributes only 1, so a single outlier dominates the total. MAE grows only linearly with the error and is therefore less sensitive to outliers.

Question 10: Real-World Applications of Supervised Learning (5 marks)

Question: *Discuss any four real-world applications of Supervised Learning. For each application, mention the type of supervised learning problem (classification or regression) and a suitable algorithm that can be used.*

Answer:

Application	Problem type	Suitable algorithm	Description
Email spam filtering	Classification	Naive Bayes / Logistic Regression / SVM	Classifies emails as spam or not spam using word features
House price prediction	Regression	Linear Regression / Random Forest Regressor	Predicts price from area, location, bedrooms, age
Medical diagnosis (e.g. disease detection)	Classification	Decision Tree / Random Forest / SVM	Predicts presence or absence of disease from symptoms and test results
Sales / demand forecasting	Regression	Gradient Boosting / Linear Regression	Predicts future sales or stock demand from past data, season, promotions
Credit card fraud / loan default detection (alternative)	Classification	XGBoost / Logistic Regression	Flags fraudulent transactions or risky borrowers

The first four rows answer the question; the fifth is an additional example.